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RL-aided cognitive ISAC: robust detection and sensing–communication trade-offs

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Abstract

This paper proposes a reinforcement learning (RL)-aided cognitive framework for massive MIMO-based integrated sensing and communication (ISAC) systems employing a uniform planar array (UPA). The focus is on enhancing radar sensing performance in environments with unknown and nonstationary disturbance characteristics. A Wald-type detector is employed for robust target detection under non-Gaussian clutter, while a SARSA-based RL algorithm enables adaptive estimation of targets' angular position in azimuth and elevation direction without prior environmental knowledge. Based on the RL-derived sensing information, a joint waveform optimization strategy is formulated to balance radar detection probability and downlink communication throughput. The resulting design provides an adaptive trade-off between detection performance and achievable sum rate through an analytically derived closed-form solution. Monte Carlo simulations demonstrate that the proposed cognitive ISAC framework achieves significantly improved detection probability compared to orthogonal and non-RL adaptive baselines, while maintaining competitive communication performance. These results underline the potential of RL-assisted sensing for robust and spectrum-efficient ISAC in next-generation wireless networks.

Keywords: Cognitive ISAC, Radar, Reinforcement learning, MIMO, Robust detection

1 Introduction

Integrated Sensing and Communication (ISAC) [1] has emerged as a central paradigm for sixth-generation (6 G) wireless systems [2], unifying sensing and communication within a shared framework of spectrum, signal processing, and hardware resources. By breaking the traditional separation between these two functionalities, ISAC not only improves spectral efficiency but also enables wireless networks to perceive, interpret, and interact intelligently with their physical environment. This dual capability supports applications such as autonomous transport, smart cities, industrial automation, and next-generation wireless monitoring [3]. Within this broader vision, radar technology serves as a cornerstone by providing high-resolution environmental awareness. When radar sensing is jointly designed with communication links, the resulting system can achieve real-time situational understanding, enhance network intelligence, and facilitate informed decision-making across distributed and autonomous infrastructures [4].

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A promising research direction within advanced radar systems is *Cognitive Radar* (CR), which extends conventional radar paradigms by embedding intelligence, adaptability, and environmental awareness into the sensing process [5, 6]. Unlike traditional radars that operate under fixed transmission parameters, CR employs the perception–action cycle, wherein the system continuously senses the environment, interprets received observations (e.g., target detections, clutter, or interference), and adaptively adjusts its transmission and reception strategies. This closed-loop interaction between transmitter and receiver enables real-time optimization of waveform design, pulse repetition frequency, and beamforming direction. Consequently, CR demonstrates superior detection and tracking performance in dynamic, cluttered, and interference-dominated environments, where static radar configurations often fail to maintain reliability [7–9]. Furthermore, CR represents a step toward self-optimizing sensing systems that can autonomously adapt to environmental uncertainty without human intervention.

Building upon these adaptive principles, *Reinforcement Learning* (RL) has recently gained attention as a powerful enabler for achieving fully autonomous radar operation [10]. RL provides a data-driven framework in which the radar learns optimal sensing and transmission policies through interaction with its environment. By optimizing a reward function aligned with radar performance objectives, such as target detection accuracy, interference mitigation, and energy efficiency, the RL agent iteratively refines its decision-making process through a balance between exploration and exploitation [11]. This paradigm eliminates the need for prior statistical knowledge of the environment, making RL particularly suitable for cognitive radar applications characterized by nonstationary and uncertain conditions. Recent studies demonstrate that RL-driven beamforming strategies for massive MIMO CR systems can achieve robust multi-target detection without explicit disturbance modeling, outperforming both conventional and adaptive non-learning baselines, particularly in low signal-to-noise ratio (SNR) and dynamically varying environments [10, 12–16].

Extending RL methodologies to the ISAC domain, recent research has investigated joint waveform design approaches that simultaneously meet both sensing and communication requirements within a unified signal framework. Unlike conventional techniques that sequentially adapt radar or communication waveforms, these approaches aim to co-optimize both functionalities from a shared resource perspective [17]. For instance, in [18], an RL-based framework for joint waveform optimization in cognitive massive MIMO ISAC systems was proposed, employing a uniform linear array (ULA) ISAC MIMO base station (BS) to enable efficient spectrum sharing without prior environmental knowledge. By iteratively probing the surroundings, updating the waveform, and balancing sensing and communication objectives, the system achieved enhanced target detection accuracy while preserving communication performance. Simulation results demonstrated consistent superiority over traditional orthogonal waveform designs, particularly under low-SNR conditions. However, the study in [18] did not explore the influence of varying communication user density on sensing performance, nor did it analyze the impact of different trade-off parameters that prioritize either sensing or communication objectives. Moreover, the analysis was limited to a ULA-based BS configuration, which restricts spatial adaptability.

To address these limitations, this work proposes an enhanced monostatic cognitive ISAC framework employing a two-dimensional (2D) uniform planar array (UPA), generalizing our previous radar-centric work [16]. Here, monostatic denotes a collocated sensing configuration in which the BS probes and observes the scene using its own transmitted signal. Accordingly, the proposed framework is restricted to monostatic ISAC. Other realizations, including bistatic or distributed-MIMO sensing, involve geographically separated nodes contributing observations to the sensing task and are therefore not considered in this paper [19–21]. In contrast to existing monostatic cognitive ISAC studies, the proposed design jointly exploits spatial diversity in both azimuth and elevation to enhance adaptive beamforming and sensing resolution. In this work, a rigorous mathematical formulation of the ISAC system is developed, wherein a Wald-type detector [22] is utilized to achieve robust target detection in massive MIMO systems operating under uncertain and dynamic environments. Building on this formulation, an RL-based optimization strategy is introduced to enable adaptive detection and waveform design in scenarios with limited prior knowledge of the environment. Based on the optimized sensing waveform, a trade-off waveform is further constructed to balance sensing accuracy and communication throughput effectively, where sensing accuracy refers to target detection performance, quantified by the probability of detection. Extensive Monte Carlo simulations are performed to evaluate the proposed framework, demonstrating that the cognitive ISAC design consistently outperforms conventional benchmark approaches, including orthogonal waveform schemes and adaptive non-RL (NRL) methods, by achieving superior sensing performance while maintaining competitive communication rates.

Although the proposed framework builds on established components such as Wald-type detection [22], analytical ISAC trade-off optimization [17], and tabular RL [11], its novelty lies in unifying these components into a closed-loop cognitive MIMO ISAC architecture that operates over a two-dimensional angular domain and adapts online under unknown, nonstationary, and heavy-tailed environments. In contrast to prior works that typically assume known sensing covariances, simplified one-dimensional spatial models, or stationary disturbance statistics, the proposed framework jointly couples robust detection, RL-based 2D target-support learning, and adaptive waveform design within a single cognitive perception-action loop. Specifically, the main contributions of this paper are as follows:

- Extending beyond our conference paper [16], which considered a radar-only setup, we develop a unified ISAC framework based on a UPA that enables joint azimuth–elevation sensing in cognitive ISAC.
- In contrast to [17], where the radar covariance is assumed available a priori, we construct the transmit covariance matrix adaptively from RL-inferred 2D target-location estimates, thus closing the loop between environment perception and ISAC waveform optimization.
- We integrate a Wald-type detector [22] into the cognitive loop as a CFAR-capable, single-snapshot robust sensing front-end under uncertain and heavy-tailed disturbance, modeled through a 2D AR clutter field with Student- t innovations. This

makes RL-driven sensing viable in practical ISAC settings with limited homogeneous training data and nonstationary clutter statistics.

- Beyond algorithm design, we provide a systematic characterization of how the proposed RL-driven adaptation reshapes the sensing–communication trade-off by quantifying its impact on detection probability and downlink sum rate across trade-off factors, transmit SNR, user density, and dynamic multi-target scenarios.

The remainder of this paper is organized as follows. Section 2 presents the detailed system model, outlines the problem formulation, and describes the proposed cognitive ISAC framework, including the RL-based decision-making mechanism. Section 3 provides a comprehensive analysis of simulation results, emphasizing the performance trade-offs between sensing accuracy and communication quality across different operational conditions, including varying SNR levels and target mobility patterns. Finally, Sect. 4 concludes the paper by summarizing the key findings, highlighting the contributions in the context of state-of-the-art ISAC research.

2 Methods

In this section, we provide a comprehensive description of the methodologies adopted in this work. We begin by formulating the ISAC system model, detailing the signal structures for both radar sensing and data transmission functionalities. Subsequently, we introduce the proposed cognitive ISAC framework, which enables the joint optimization of waveform parameters to achieve an adaptive balance between sensing accuracy and communication reliability. The formulation integrates both domains under a unified optimization criterion, ensuring efficient resource utilization and mutual coexistence. Finally, we describe the RL-based adaptation mechanism developed to enhance target detection performance under dynamic operating conditions.

2.1 System model

We consider a MIMO ISAC BS that simultaneously performs dual functionalities of wireless communication and radar sensing, as depicted in Fig. 1. Specifically, the BS transmits information-bearing symbols to K single-antenna downlink users, while concurrently operating as a monostatic, colocated cognitive radar [10, 22, 23]. In this configuration, a unified transmit waveform is employed for both tasks: conveying data to the communication users and probing the surrounding environment for target detection. This joint utilization of the transmitted signal enables efficient spectrum sharing and hardware reuse, thereby enhancing the overall system performance and resource efficiency. More precisely, the resource efficiency in the considered ISAC model stems from sharing the transmit waveform, transmit aperture, spectrum, and BS platform between communication and sensing. At the same time, monostatic sensing during downlink transmission requires that the BS illuminates the scene and collects the corresponding echoes simultaneously. Therefore, a receive array is required to acquire the radar echoes in parallel with the shared transmit aperture, under the assumed full-duplex operation with sufficient Tx–Rx isolation and self-interference cancellation [24–26].

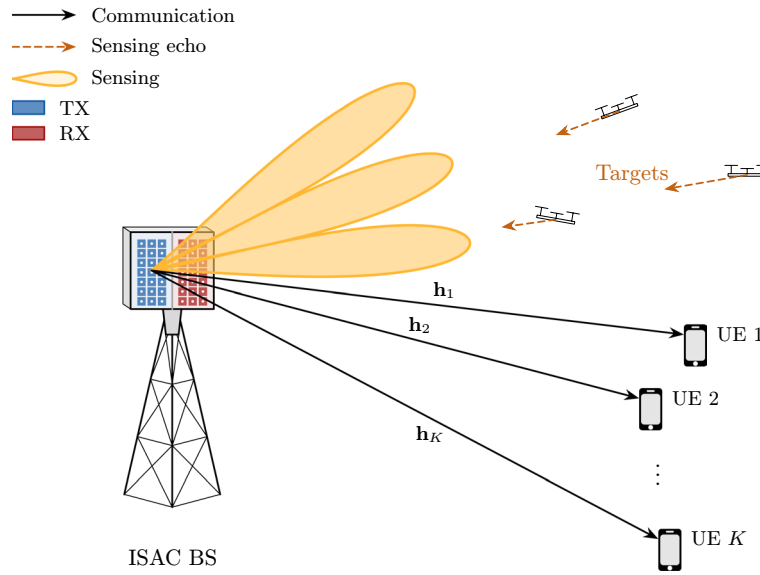


Fig. 1 ISAC BS system serving K single-antenna users while operating as a monostatic cognitive radar

2.1.1 Radar signal model

Consider a colocated MIMO radar equipped with N_t transmit and N_r receive antennas, yielding a total of $N = N_t N_r$ virtual channels. Both arrays are modeled as UPAs with half-wavelength element spacing [16]. The target is located at elevation angle θ and azimuth angle ϕ .

At the l -th signaling interval ($l = 1, \dots, L$), the transmitted narrowband waveform vector $\mathbf{x}^l \in \mathbb{C}^{N_t}$ is emitted, where L denotes the code length [27]. The corresponding received baseband echo is given by

$$\mathbf{y}_s^l = \alpha \mathbf{a}_r(\theta, \phi) \mathbf{a}_t(\theta, \phi)^T \mathbf{x}^l + \mathbf{c}_s^l, \quad (1)$$

where $\mathbf{y}_s^l \in \mathbb{C}^{N_r}$ is the received signal vector, $\alpha \in \mathbb{C}$ represents the target's complex scattering coefficient accounting for the radar cross section and two-way propagation loss [28], and $\mathbf{c}_s^l$ denotes the additive clutter-plus-noise term, later modeled as a AR process with covariance Γ .

The transmit and receive steering vectors $\mathbf{a}_t(\theta, \phi) \in \mathbb{C}^{N_t}$ and $\mathbf{a}_r(\theta, \phi) \in \mathbb{C}^{N_r}$ follow the separable UPA structure [29]

$$\mathbf{a}_i(\theta, \phi) = \mathbf{a}_{x,i}(\theta, \phi) \otimes \mathbf{a}_{y,i}(\theta, \phi), \quad i \in \{t, r\}. \quad (2)$$

where $\otimes$ denotes the Kronecker product, and $\mathbf{a}_x(\theta, \phi)$ and $\mathbf{a}_y(\theta, \phi)$ represent the steering responses along the horizontal and vertical array dimensions, respectively.

By stacking the L received snapshots into a matrix form, the overall signal model becomes

$$\mathbf{Y}_s = \alpha \mathbf{a}_r(\theta, \phi) \mathbf{a}_t(\theta, \phi)^T \mathbf{X} + \mathbf{C}_s, \quad (3)$$

where $\mathbf{X} = [\mathbf{x}^1, \dots, \mathbf{x}^L] \in \mathbb{C}^{N_t \times L}$ contains the transmitted signals, and $\mathbf{Y}_s, \mathbf{C}_s \in \mathbb{C}^{N_r \times L}$ denote the received and disturbance matrices, respectively.

After correlating the received signal with the transmitted waveform matrix at the receiver and vectorizing the result, (3) can be expressed in terms of the transmit covariance matrix $\mathbf{R} = \frac{1}{L} \mathbf{X} \mathbf{X}^H$ [30] as

$$\mathbf{y}_s = \alpha \text{vec} \left(\mathbf{a}_r(\theta, \phi) \mathbf{a}_t(\theta, \phi)^T \mathbf{R} \right) + \mathbf{c}_s, \quad (4)$$

where $\mathbf{y}_s, \mathbf{c}_s \in \mathbb{C}^{N_r N_t}$ are the vectorized received signal and disturbance terms, respectively, and $\mathbf{R} \in \mathbb{C}^{N_t \times N_t}$ is the transmit covariance matrix. Using the identity $\text{vec}(\mathbf{u} \mathbf{v}^T) = \mathbf{v} \otimes \mathbf{u}$, define

$$\mathbf{h} = (\mathbf{R}^T \mathbf{a}_t(\theta, \phi)) \otimes \mathbf{a}_r(\theta, \phi) \in \mathbb{C}^{N_r N_t}. \quad (5)$$

Then the model reduces to

$$\mathbf{y}_s = \alpha \mathbf{h} + \mathbf{c}_s. \quad (6)$$

We assume a fully-digital receive architecture where the baseband signal vector across the antenna elements is available for processing. This configuration enables the estimation of the spatial disturbance covariance matrix required by the Wald-type detector [22] later.

For spatial processing, the radar field of view is discretized into $M_x \times M_y$ angular bins, as illustrated in Fig. 2.

The underlying spatial frequencies are defined as

$$\nu_x = \frac{1}{2} \sin(\theta) \cos(\phi), \quad \nu_y = \frac{1}{2} \sin(\theta) \sin(\phi), \quad (7)$$

assuming half-wavelength inter-element spacing. Sampling these spatial frequencies on a discrete grid yields

$$\nu_x^{(i)}, \quad i = 1, \dots, M_x, \quad \nu_y^{(j)}, \quad j = 1, \dots, M_y. \quad (8)$$

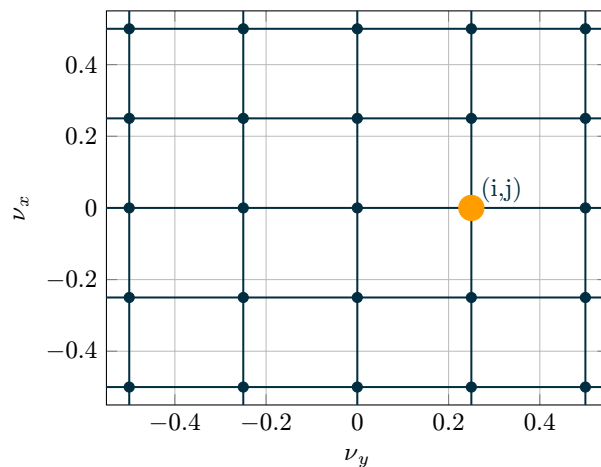


Fig. 2 Discrete spatial-frequency grid for $M_x = M_y = 5$. The spatial-frequency plane (ν_x, ν_y) is sampled on a rectangular grid. Each grid point $(\nu_x^{(i)}, \nu_y^{(j)})$ corresponds to a spatial bin indexed by (i, j) and mapped to a spatial location $m \in \mathcal{M}$

Each bin (i, j) corresponds to a unique spatial location $m \in \mathcal{M}$, where $|\mathcal{M}| = M_x M_y$.

To simplify notation, we omit the subscript s in the following so as to obtain a cleaner formulation. Over P pulses indexed by $p = 1, \dots, P$, the signal model for the m -th spatial bin is given by

$$\mathbf{y}_{p,m} = \alpha_{p,m} \mathbf{h}_{p,m} + \mathbf{c}_{p,m}, \quad (9)$$

where $\alpha_{p,m}$ denotes the complex target amplitude and $\mathbf{c}_{p,m}$ represents clutter-plus-noise at that bin and pulse.

The binary detection problem for each spatial bin m and pulse p can then be formulated as

$$\mathcal{H}_0 : \mathbf{y}_{p,m} = \mathbf{c}_{p,m}, \quad (10)$$

$$\mathcal{H}_1 : \mathbf{y}_{p,m} = \alpha_{p,m} \mathbf{h}_{p,m} + \mathbf{c}_{p,m}. \quad (11)$$

A decision statistic $\Lambda_{p,m}$ is compared to a threshold η according to

$$\Lambda_{p,m} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\gtrless}} \eta, \quad (12)$$

where $\eta = -2 \ln(P_{FA})$ ensures a prescribed false-alarm probability. Employing a Wald-type detector [22], the test statistic is defined as

$$\Lambda_{p,m} = \frac{2|\mathbf{h}_{p,m}^H \mathbf{y}_{p,m}|^2}{\mathbf{h}_{p,m}^H \hat{\Gamma}_{p,m} \mathbf{h}_{p,m}}, \quad (13)$$

where $\hat{\Gamma}_{p,m}$ denotes the estimated clutter-plus-noise covariance matrix [10]. Further details on the estimation of $\Gamma_{p,m}$ are provided in Appendix A. Unlike conventional space–time adaptive processing (STAP) filter, which requires a large set of homogeneous secondary data for clutter covariance estimation, the adopted Wald-type detector operates reliably under limited or non-homogeneous observations. This property is essential in cognitive ISAC scenarios, where the clutter statistics may vary rapidly and only a single-snapshot per pulse is available. Moreover, the Wald detector exhibits robustness to non-Gaussian and heavy-tailed disturbances and enables analytical tractability for RL-based adaptation.

As mentioned earlier, this work assumes digital beamforming. In hybrid analog/digital beamforming architectures, the proposed detector could, in principle, be applied to the post-combining observations, yielding a lower-dimensional covariance estimate. This would lead to a poorer estimate of the covariance [10, 22]. However, the treatment of hybrid beamforming architectures lies beyond the scope of the present work and is therefore not considered.

2.1.2 Communication signal model

The dual-functional radar-communication system must also guarantee reliable downlink performance. Specifically, the transmitted signal $\mathbf{X}$ simultaneously serves as both a radar probing waveform and a communication data carrier [17]. This inherent coupling necessitates a careful joint design to satisfy radar beam pattern requirements while maintaining

communication quality-of-service (QoS) criteria, such as the achievable sum-rate and symbol error rate (SER) [17].

The received downlink signal at K single-antenna users can be expressed as

$$\mathbf{Y}_c = \mathbf{H}\mathbf{X} + \mathbf{N}, \quad (14)$$

where $\mathbf{H} \in \mathbb{C}^{K \times N_t}$ denotes the flat Rayleigh fading channel matrix and $\mathbf{N} \in \mathbb{C}^{K \times L}$ is additive white Gaussian noise (AWGN). $\mathbf{X}$ represents the dual-use waveform transmitted for both radar and communication. The flat Rayleigh channel is adopted for analytical tractability and to keep the focus on the proposed learning-aided ISAC waveform adaptation [31]. The formulation extends to Rician channels by using the appropriate model for $\mathbf{H}$, and to frequency-selective channels by applying the design per subcarrier.

The channel $\mathbf{H}$ is assumed to be perfectly known at the transmitter and to remain constant within one radar pulse or one communication frame. Perfect knowledge of $\mathbf{H}$ is assumed to isolate the sensing–communication coupling and obtain a clean analytical characterization of the waveform trade-off without confounding the results with channel acquisition and estimation effects. The resulting formulation therefore characterizes the achievable performance under ideal channel state information (CSI) [31]. The block-fading assumption is justified whenever the frame duration does not exceed the channel coherence time, i.e.,

$$T_{\text{block}} = L \cdot T_s \ll T_c,$$

where T_{block} denotes the block duration, T_s denotes the symbol duration and T_c the channel coherence time.

Let $\mathbf{S} \in \mathbb{C}^{K \times L}$ denote the target constellation matrix corresponding to the intended information symbols for all users. The received signal can equivalently be decomposed as

$$\mathbf{Y}_c = \mathbf{S} + \underbrace{(\mathbf{H}\mathbf{X} - \mathbf{S})}_{\text{MUI}} + \mathbf{N}, \quad (15)$$

where the underbraced term quantifies the multi-user interference (MUI) resulting from imperfect spatial separation among users. The overall MUI energy is defined as

$$P_{\text{MUI}} = \|\mathbf{H}\mathbf{X} - \mathbf{S}\|_F^2, \quad (16)$$

which directly influences the achievable downlink throughput and user fairness.

For the k -th user, the instantaneous signal-to-interference-plus-noise ratio (SINR) per frame is given by

$$\gamma_k = \frac{\mathbb{E}[|s_{k,j}|^2]}{\mathbb{E}[|h_k^T x_j - s_{k,j}|^2] + N_0}, \quad (17)$$

where $s_{k,j}$ denotes the (k, j) -th entry of $\mathbf{S}$, h_k^T is the k -th row of $\mathbf{H}$, and x_j is the j -th column of $\mathbf{X}$. Consequently, the overall achievable downlink sum-rate can be expressed as

$$R = \sum_{k=1}^K \log_2(1 + \gamma_k). \quad (18)$$

2.2 Cognitive ISAC for sensing–communication trade-offs

The performance objectives of communication and radar sensing systems are fundamentally different, reflecting their distinct operational goals. In communication systems, the primary objective is to transmit information reliably and efficiently from the source to the intended receiver. Conversely, radar sensing is not concerned with maximizing information transfer, but with extracting relevant information from the echoes of the transmitted signals. In essence, communication systems embed the information of interest within the transmitted waveform itself, whereas in radar systems, the desired information is contained within the received reflections.

This fundamental distinction motivates the design challenge in ISAC systems: A single waveform must serve two purposes simultaneously. From the system models previously introduced, one approach is to formulate waveform design as an optimization problem that balances dual objectives: (i) minimizing the MUI power for reliable communication, and (ii) achieving a desired beampattern for effective radar detection.

A widely studied formulation for the joint radar-communication trade-off was introduced in [17] and can be expressed as

$$\mathcal{P}_0 : \min_{\mathbf{X}} \quad \rho \|\mathbf{H}\mathbf{X} - \mathbf{S}\|_F^2 + (1 - \rho) \|\mathbf{X} - \mathbf{X}_0\|_F^2 \quad (19)$$

$$\text{s.t.} \quad \frac{1}{L} \|\mathbf{X}\|_F^2 = P_T. \quad (19a)$$

The parameter $0 \leq \rho \leq 1$ acts as a trade-off factor that adjusts the priority between the two system objectives. The first term in (19) penalizes MUI, thereby improving communication reliability, while the second term encourages similarity to a reference waveform $\mathbf{X}_0$, thus preserving radar functionality.

To facilitate analysis, the composite objective function can be reformulated as

$$\rho \|\mathbf{H}\mathbf{X} - \mathbf{S}\|_F^2 + (1 - \rho) \|\mathbf{X} - \mathbf{X}_0\|_F^2 = \|\mathbf{A}\mathbf{X} - \mathbf{B}\|_F^2, \quad (20)$$

where

$$\mathbf{A} = \begin{bmatrix} \sqrt{\rho} \mathbf{H} \\ \sqrt{1-\rho} \mathbf{I}_{N_t} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \sqrt{\rho} \mathbf{S} \\ \sqrt{1-\rho} \mathbf{X}_0 \end{bmatrix}. \quad (21)$$

By applying the reformulation in (20), the optimization task can be expressed as

$$\begin{aligned} \mathcal{P}_{0.1} : \quad & \min_{\mathbf{X}} \quad \|\mathbf{A}\mathbf{X} - \mathbf{B}\|_F^2 \\ & \text{s.t.} \quad \|\mathbf{X}\|_F^2 = LP_T. \end{aligned} \quad (22)$$

Problem $\mathcal{P}_{0.1}$ belongs to the class of quadratically constrained quadratic programs (QCQPs). Since the quadratic equality constraint renders the problem non-convex, solving it exactly is computationally challenging. A widely adopted approach to address this difficulty is to employ the semidefinite relaxation (SDR) framework, which reformulates the QCQP into a semidefinite program (SDP). This convex relaxation allows the use of efficient numerical solvers while still providing practically useful solutions. The optimal solution for this problem is derived in [17] as

$$\mathbf{X}_{opt} = (\mathbf{A}^H \mathbf{A} + \lambda_{opt} \mathbf{I}_{N_t})^\dagger (\mathbf{A}^H \mathbf{B}), \quad (23)$$

where λ_{opt} is the optimal dual point and eigenvalue of $\mathbf{Q} = \mathbf{A}^H \mathbf{A}$, which can be obtained by simple line search methods, e.g., Golden-section search. More details on how to derive this solution and the algorithm can be found in Appendix B.

Having derived the optimal waveform for the considered ISAC system, it remains necessary to construct a reference waveform, denoted as $\mathbf{X}_0$, which is optimal for target detection. To this end, we formulate the following optimization problem

$$\begin{aligned} \mathcal{P}_1 : \quad & \min_{\mathbf{X}_0} \|\mathbf{H}\mathbf{X}_0 - \mathbf{S}\|_F^2 \\ & \text{s.t.} \quad \frac{1}{L} \mathbf{X}_0 \mathbf{X}_0^H = \mathbf{R}_d, \end{aligned} \quad (24)$$

where $\mathbf{R}_d$ represents the desired transmit covariance matrix, which corresponds to a target beampattern. The objective function minimizes the MUI while ensuring that the resulting waveform adheres to the specified covariance constraint $\mathbf{R}_d$.

A closed-form solution to problem $\mathcal{P}_1$ is provided in [17] as

$$\mathbf{X}_{0,opt} = \sqrt{L} \mathbf{F} \mathbf{U} \mathbf{I}_{N \times L} \mathbf{V}^H, \quad (25)$$

where $\mathbf{R}_d = \mathbf{F} \mathbf{F}^H$ and the singular value decomposition (SVD) $\mathbf{U} \mathbf{\Sigma} \mathbf{V}^H = \mathbf{F}^H \mathbf{H}^H \mathbf{S}$ holds. The detailed derivation of this closed-form solution is provided in Appendix C.

In [17], the optimization problem in $\mathcal{P}_1$ was addressed under the assumption that the transmit covariance matrix $\mathbf{R}_d$ is perfectly known a priori. However, this assumption is unrealistic, since both the number of targets and their directions of arrival are typically unknown prior to the sensing process. In this work, we employ RL to estimate the targets' angular position in azimuth and elevation direction. Based on these estimates, the transmit covariance matrix $\mathbf{R}_d$ is subsequently designed by solving the following optimization problem [32]

$$\mathcal{P}_2 : \quad \max_{\mathbf{R}_d} \text{tr}(\mathbf{R}_d \mathbf{\Theta}) \quad (26)$$

$$\text{s.t.} \quad \text{tr}(\mathbf{R}_d) = P_T, \quad (27)$$

$$\mathbf{R}_d \succeq 0, \quad (28)$$

where

$$\mathbf{\Theta} = \sum_{b=1}^T \mathbf{a}_t(\theta_b, \phi_b) \mathbf{a}_t^H(\theta_b, \phi_b) \quad (29)$$

where T denotes the number of targets and $\mathbf{a}_t(\theta_b, \phi_b)$ the transmit steering vector of the target b . The closed-form solution of problem $\mathcal{P}_2$ is given by

$$\mathbf{R}_{d,opt} = P_T \mathbf{u} \mathbf{u}^*, \quad (30)$$

where $\mathbf{u}$ denotes the unit-norm eigenvector of $\hat{\mathbf{B}}$ corresponding to its largest eigenvalue $\lambda_{\max}(\hat{\mathbf{B}})$ [32, 33]. Hence, problem $\mathcal{P}_2$ maximizes the total transmitted power over the estimated target locations.

Naturally, the angular positions of the targets are not known *a priori*. Therefore, a RL framework is developed in the following to learn the target locations and calculate the transmit covariance matrix with $\mathcal{P}_2$.

2.3 Reinforcement learning framework

RL is a branch of machine learning in which an intelligent agent gradually improves its performance by interacting with an environment [11]. Unlike supervised learning, where the correct output is provided for each example, in RL the agent must discover successful strategies on its own through repeated trials and feedback. This process can be conceptualized as an iterative cycle of trial and error: The agent takes an action, observes how the environment responds, and then adjusts its future behavior in order to achieve better outcomes. Over time, this iterative process enables the agent to learn a policy π that is, a set of rules that guide which action should be taken in a given situation. The overall RL mechanism applied in our ISAC scenario is illustrated in Fig. 3.

In the context of our work, the ISAC system plays the role of the agent. Compared with a conventional direction of arrival (DoA) estimator, the proposed framework does not merely estimate target directions, but closes the loop between detection, learning, and waveform adaptation in a cognitive ISAC setting. This enables robust identification of target-bearing angular bins under uncertain disturbance conditions and uses this information to adapt the dual-functional transmit covariance/waveform, thereby improving sensing performance while accounting for communication requirements. The ISAC agent receives continuous feedback from the surrounding environment, evaluates this feedback, and then decides how to act next. To formalize the learning mechanism, we employ the SARSA algorithm, which is a widely used RL method [11]. The name “SARSA” comes from the sequence of elements involved in each learning step: *State, Action, Reward, next State, and next Action*. At each step, the agent observes its current state s_p , chooses an action a_p , and then receives a reward r_{p+1} from the environment. After transitioning to a new state s_{p+1} , the agent selects its next action a_{p+1} . This entire sequence is then used to update the agent’s knowledge of the environment.

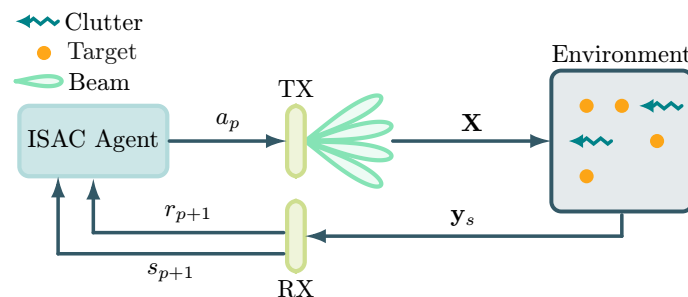


Fig. 3 RL framework for the SARSA-based cognitive ISAC

The agent's decision-making is guided by a function known as the Q-value, which measures the long-term benefit of taking a particular action in a given state under a policy π [11]. Mathematically, the SARSA update rule is expressed as

$$Q(s_p, a_p) \leftarrow Q(s_p, a_p) + \alpha [r_{p+1} + \gamma Q(s_{p+1}, a_{p+1}) - Q(s_p, a_p)], \quad (31)$$

where $\alpha \in [0, 1]$ is the learning rate that controls how strongly new information influences the Q-values, and $\gamma \in [0, 1]$ is the discount factor that determines how much importance is given to future rewards compared to immediate ones.

In our ISAC setup, SARSA is used to efficiently explore the environment and detect the targets under unknown conditions. The knowledge acquired during learning is stored in a state–action value matrix, denoted by $\mathbf{Q}$. Its size depends on the dimensions of the state and action spaces, which will be introduced later. This matrix is continuously updated so that the joint waveform of the ISAC system can adapt dynamically to changes in the environment.

The next subsection presents the specific components of the learning process for the cognitive ISAC system: the state, the action, reward and the SARSA-based algorithm.

2.3.1 States and actions

The state observed at each pulse sent at the time p is represented by state s_p . This state provides a representation of the sensing environment at time p , encapsulating the detection outcome of the radar sensing process. In other words, s_p carries information about what the system has detected. Specifically, the state is the total number of angular bins whose corresponding test statistics exceed a predefined detection threshold η , thereby indicating the presence of a possible target. To formalize the detection outcome, we introduce a binary indicator for each bin $m \in \mathcal{M}$. This indicator captures whether or not a detection is declared in bin m at pulse p

$$\hat{\Lambda}_{p,m} = \begin{cases} 1, & \text{if } \Lambda_{p,m} > \eta, \\ 0, & \text{otherwise.} \end{cases} \quad (32)$$

Here, $\hat{\Lambda}_{p,m} = 1$ indicates that the received test statistic $\Lambda_{p,m}$ surpasses the threshold η , and thus, a detection is declared for bin m . Conversely, $\hat{\Lambda}_{p,m} = 0$ denotes that no detection is made in that bin. Summing over all bins yields the total number of detections from the pulse sent at time step p , which is given by

$$T_p = \sum_{m \in \mathcal{M}} \hat{\Lambda}_{p,m}, \quad (33)$$

where $T_p \in \{0, 1, \dots, \tilde{T}\}$ represents the total number of detections at time step p . The parameter $\tilde{T}$ denotes the maximum number of detectable targets, thereby defining the size of the state space. In practice, this limitation ensures that the state space remains finite and manageable for the RL agent. It is theoretically bounded by the angular discretization, i.e., $T_p \in \{0, \dots, |\mathcal{M}|\}$, since at most one detection can occur per spatial bin. In practice, however, the number of reliably resolvable targets is smaller and depends

on the sensing configuration and environment (array aperture, beampattern sidelobes, grid resolution, the threshold set by P_{FA} , SNR/clutter statistics, and target separability). $\tilde{T}$ is therefore a design parameter representing the maximum target multiplicity handled by the RL agent and cap the RL state space accordingly. $\tilde{T}$ may be chosen from prior knowledge of scene occupancy or from a conservative bound based on resolvability and computational budget. Note that, choosing $\tilde{T}$ larger than necessary increases the size of the RL state–action space and thus the Q-table and exploration burden. For implementation purposes, the state is defined as $s_p = T_p + 1$ and the state space is defined as $S := \{1, \dots, \tilde{T} + 1\}$. The action chosen at step p is represented by a_p .

The action is chosen from the action space $\mathcal{A} \in \{\Theta_0, \Theta_1, \dots, \Theta_{\tilde{T}}\}$, and each Θ_j corresponds to the set of indices associated with the top- j values of $\Lambda_{p,m}$. Thus, Θ_j encodes the set of candidate bins that are prioritized for sensing at the pulse sent at time step p . Based on the chosen action, the optimization procedure unfolds in multiple stages. First, problem $\mathcal{P}_2$ is solved using the candidate bins specified by Θ_p , yielding the optimal covariance matrix $\mathbf{R}_d$ that governs the sensing process. This intermediate result is then employed in problem $\mathcal{P}_1$, where the optimal waveform X_0 is determined to minimize MUI in the communication domain. Finally, the solution of $\mathcal{P}_1$ is used to optimize problem $\mathcal{P}_0$. The final output of this sequential optimization pipeline is an optimized waveform, which balances sensing accuracy and communication quality, and is transmitted in the subsequent pulse. In this way, the RL agent continuously adapts its actions across pulses, enabling a dynamic and context-aware allocation of resources between radar sensing and wireless communication.

2.3.2 Reward

Having defined the ISAC agent's action and state spaces, we now focus on designing an appropriate reward function that guides its behavior toward accurate target detection. The reward function plays a key role in shaping the agent's learning objective. For radar sensing, a natural performance measure is the probability of detection P_D , balanced against the probability of false alarm P_{FA} . Intuitively, the reward should encourage correct detections in bins containing targets while penalizing false detections in empty bins.

Let $\mathcal{T}$ denote the set of bins that contain a target, and let $\mathcal{N}$ be its complement. The set $\mathcal{T}$ is defined based on the detector output as $\mathcal{T} \triangleq \{m \in \mathcal{M} : \Lambda_{p,m} > \eta\}$, i.e., the set of bins whose test statistic exceeds the threshold (and for which a detection is therefore declared), while $\mathcal{N} \triangleq \mathcal{M} \setminus \mathcal{T}$. Following [22], the theoretical detection performance at bin m can be characterized by an asymptotic detection probability $\hat{P}_{D,(p,m)}$, approximated by a first-order Marcum Q-function [34]

$$\hat{P}_{D,(p,m)} \stackrel{N \rightarrow \infty}{\approx} Q_1(\sqrt{\kappa_{p,m}}, \sqrt{\eta}), \quad (34)$$

where η is the detection threshold, and $\kappa_{p,m}$ is given by

$$\kappa_{p,m} = \frac{2 |\hat{\alpha}_{p,m}|^2 \|\mathbf{h}_{p,m}\|^4}{\mathbf{h}_{p,m}^H \hat{\Gamma}_m \mathbf{h}_{p,m}}, \quad \hat{\alpha}_{p,m} = \frac{\mathbf{h}_{p,m}^H \mathbf{y}_{p,m}}{\|\mathbf{h}_{p,m}\|^2}. \quad (35)$$

Based on this, the reward at time step p is defined as

$$r_p = \sum_{m \in \mathcal{T}} \hat{P}_{D,(p,m)} - \sum_{m \in \mathcal{N}} \hat{P}_{D,(p,m)}, \quad (36)$$

which encourages the agent to correctly identify targets while penalizing false alarms.

2.3.3 SARSA-based algorithm

In the previous sections, we defined the key components of our RL framework, including states, actions, and the reward function. Building upon these foundations, we next introduce the SARSA-based algorithm. SARSA is a model-free, on-policy RL method known for inherently balancing exploration and exploitation, making it particularly advantageous in real-time scenarios where environmental conditions are often nonstationary [11]. Moreover, its simplicity in implementation and adaptive nature contribute to robust performance even under dynamic conditions. Central to the SARSA algorithm is the Q -function, as defined before. This function approximates the long-term return, defined as the expected cumulative discounted reward, that an agent accrues by taking action a in state s and subsequently following the current policy. The learning mechanism involves iteratively updating the estimate $Q(s, a)$ using the observed tuple $\{s_p, a_p, r_{p+1}, s_{p+1}, a_{p+1}\}$ [11].

To achieve robust performance of the SARSA algorithm, an ε -greedy policy is utilized to balance the trade-off between exploration and exploitation. Under this scheme, the agent primarily selects the optimal action based on the current Q -values, while occasionally exploring alternative actions with probability ε . This mechanism allows the system to adapt to environmental variations such as fluctuations in the number of detected targets, SNR, and angular positions. In this study, the quasi ε -greedy policy with target recovery, as proposed in [13], is adopted. The policy is formally expressed as

$$a_{p+1}^* = \begin{cases} \arg \max_{a \in \mathcal{A}} Q(s_p, a), & m_{p+1} < m_p, \\ \begin{cases} \text{a random action from } \mathcal{A}', & \text{w.p. } \varepsilon, \\ \arg \max_{a \in \mathcal{A}} Q(s_{p+1}, a), & \text{w.p. } 1 - \varepsilon, \end{cases} & m_{p+1} \geq m_p, \end{cases} \quad (37)$$

where $\mathcal{A}'$ denotes the subset of actions corresponding to a higher number of angular bins than those in the current state. According to this formulation, when the number of detected targets at time step $p + 1$ is equal to or greater than that at step p , the radar employs the exploratory strategy defined for the condition $m_{p+1} \geq m_p$. This prevents the system, during exploration, from concentrating its power on fewer angular bins than required for the number of potential targets. Conversely, when the number of detected targets decreases, the policy enforces a recovery phase by selecting the greedy action associated with the previous state s_p .

Building on these foundational principles, Algorithm 1 details the complete SARSA-based RL framework for ISAC. The algorithm integrates the SARSA methodology to iteratively optimize the waveform and facilitate a trade-off between robust detection decisions and high communication rates.

Algorithm 1 SARSA-Based RL Algorithm for Cognitive ISAC

Initialize $p \leftarrow 0$, $\mathbf{Q} \leftarrow \mathbf{0}_{(|\mathcal{S}|) \times |\mathcal{A}|}$, $s_0 \leftarrow 1$, $a_0 \leftarrow 1$, and $\mathbf{X}_0 \leftarrow \mathbf{I}$.
repeat for each time step p :
 1: Take action a_p by transmitting the waveform $\mathbf{X}_p$
 2: $\mathbf{y}_{p,m} \leftarrow$ acquired received signal $\forall m$
 3: $s_{p+1} \leftarrow$ result from (33), $r_{p+1} \leftarrow$ result in (36)
 4: $a_{p+1} \leftarrow$ as in (37), identify Θ_{p+1}
 5: $\mathbf{R}_{d,p+1} \leftarrow$ solution $\mathcal{P}_2$
 6: $\mathbf{X}_{0,p+1} \leftarrow$ solution in $\mathcal{P}_1$
 7: $\mathbf{X}_{p+1} \leftarrow$ solution in $\mathcal{P}_0$
 8: Update $Q(s_p, a_p)$ using (31), $s_p \leftarrow s_{p+1}$, $a_p \leftarrow a_{p+1}$.
until the observation period P terminates.

3 Results and discussion

The performance of the proposed RL framework is assessed through a series of simulation experiments. The following subsection outlines the simulation setup, including all parameter configurations and the adopted disturbance model. Subsequently, various simulation scenarios are examined and discussed, beginning with stationary targets with fixed angular positions under different numbers of communication users, and extending to dynamic environments characterized by target appearance, disappearance, and mobility.

3.1 Parameters

In our simulations, a massive MIMO BS is considered, which transmits messages of length $L = 30$ [17] to K users over a channel matrix $\mathbf{H}$ with elements $H_{i,j} \in \mathcal{CN}(0, 1)$. The channel is assumed to be perfectly estimated. The desired symbols in $\mathbf{S}$ are drawn from a quadrature phase-shift keying (QPSK) constellation with unit average power. QPSK is used as a constant-envelope baseline (as in [17]) to evaluate waveform optimization under a fixed-modulus assumption and avoid modulation-dependent amplitude variations. The framework is not restricted to QPSK; higher-order constellations are accommodated by redefining the desired symbol matrix $\mathbf{S}$.

The simulations are conducted over a spatial grid defined by $M_x = M_y = 11$, resulting in a total of 121 bins. The grid is represented in the spatial-frequency domain, where v_x and v_y take values from the discrete set $\{-0.5, -0.4, \dots, 0.5\}$. The key simulation parameters are summarized in Table 1. Unless noted otherwise, these parameters remained fixed for all subsequent experiments. The massive-MIMO-related settings (e.g., array dimensions and sensing-related operating points) and the RL hyperparameters (e.g., discount factor, exploration rate, and learning-rate choices) follow the baseline configuration and tuning principles in [10], where the numerical values were also tested across a range of alternatives and shown to provide the best performance. The communication-related parameters are adopted consistently with typical DFRC/ISAC communication simulation setups, as in [17]. Overall, these parameter choices reflect standard practice in the respective literatures and were selected to provide a representative operating

Table 1 Simulation Parameters

Parameter	Value
Probability of false alarm, P_{FA}	10^{-4}
Number of transmit antennas, N_t	100
Number of receive antennas, N_r	100
Transmit power, P_T	1
Tail heaviness parameter, μ	2
Noise variance, σ_w^2	1
Number of detectable targets, $\tilde{T}$	10
Discount factor, γ	0.8
Exploration factor ϵ	0.5
Learning rate, η	0.8

regime for evaluating the proposed method. The following simulations were each carried out using 10^3 Monte Carlo runs.

The disturbance vector $\mathbf{c}_s$ is modeled as a two-dimensional circular complex autoregressive (AR) process [35]. The process is defined as

$$c_{n_x, n_y} = \sum_{i=1}^p \sum_{j=1}^q \rho_{i,j} c_{n_x-i, n_y-j} + w_{n_x, n_y}, \quad (38)$$

where c_{n_x, n_y} denotes the disturbance component at spatial index (n_x, n_y) . The coefficients $\rho_{i,j}$ capture the spatial dependencies of the AR process, thereby characterizing its correlation structure. The term w_{n_x, n_y} represents the driving noise at each spatial location, which serves as a stochastic excitation to the system.

To more accurately model realistic radar clutter, the driving noise w_{n_x, n_y} is assumed to follow a heavy-tailed Student's t -distribution [22], given by

$$p_w(w_{n_x, n_y}) = \frac{\mu}{\sigma_w^2} \left(\frac{\mu}{\xi} \right)^\mu \left(\frac{\mu}{\xi} + \frac{|w_{n_x, n_y}|^2}{\sigma_w^2} \right)^{-(\mu+1)}, \quad (39)$$

where μ controls the heaviness of the distribution tails and σ_w^2 denotes the noise variance. This formulation effectively captures the impulsive and non-Gaussian behavior commonly encountered in high-clutter radar environments, enhancing robustness in practical detection scenarios. The values of μ and σ_w^2 used in these simulations are listed in Table 1.

The corresponding power spectral density (PSD) of the 2D AR process in the spatial-frequency domain is expressed as

$$S(v_x, v_y) = \sigma_w^2 \left| 1 - \sum_{n=1}^p \sum_{l=1}^q \rho_{n,l} e^{-j2\pi(nv_x + lv_y)} \right|^{-2}, \quad (40)$$

which characterizes the distribution of disturbance power across spatial frequencies. The AR coefficients directly influence the spectral shape of the disturbance process,

thereby determining its spatial correlation properties. The coefficient matrix employed in these simulations is given by

$$\boldsymbol{\rho} = \begin{bmatrix} 0 & 0.1 & 0.1 \\ 0.1 & 0 & 0 \\ 0.05 & 0 & 0 \end{bmatrix}. \quad (41)$$

This matrix defines the spatial correlation structure of the modeled disturbance field.

3.2 Comparison of RL-based and benchmark schemes in a stationary scene

This section presents an evaluation of the proposed system's performance in a stationary sensing environment. Here, stationary means that the targets' angular position remain fixed during the considered observation interval. It provides a baseline for evaluating sensing accuracy without the added effects of target motion. Within this scenario, four distinct targets are assumed to be present and detected by the sensing module. The spatial locations of these targets along with their corresponding SNRs are summarized in Table 2. The communication subsystem is configured to serve $K = 48$ users. The configuration corresponds to a moderately loaded massive MIMO system where the base station serves approximately half as many users as antennas. This regime is commonly considered in the literature to ensure reliable spatial multiplexing while maintaining manageable interference levels [36]. For this baseline experiment, the communication link operates at an high SNR of 12 dB. In subsequent evaluations, we perform simulations across a range of SNR values to analyze the robustness and performance trends of the proposed method under varying channel conditions. To assess the efficacy of the proposed cognitive RL-driven ISAC framework, we compare its performance against two benchmark schemes. The first employs orthogonal beam pattern selection at each time step [10], while the second represents an adaptive non-RL (NRL) approach [10, 16], wherein a beam pattern is iteratively generated to focus energy toward the spatial bins that exceed the detection threshold. For this preliminary investigation, we set the weighting factor to $\rho = 0.2$, thereby assigning a lower priority to communication performance relative to sensing accuracy.

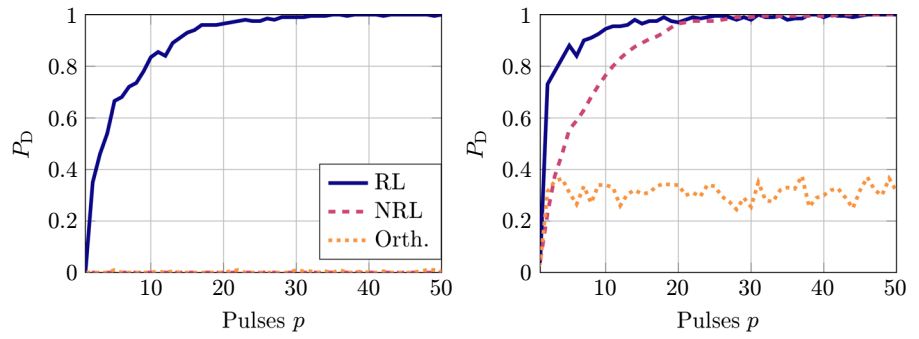
Figure 4 illustrates the temporal evolution of the detection probability for Target 1 and 2 across the considered approaches. These two targets are selected for analysis as

Table 2 Stationary Targets

Target	Time interval [1, 50]
1	$(-0.4, -0.4)^1$ -30 dB ²
2	$(0, 0)$ -25 dB
3	$(0.3, 0.1)$ -20 dB
4	$(-0.1, 0.4)$ -15 dB

¹ Target's angular position

² Target's SNR



(a) P_D over time for Target 1 (-30 dB) (b) P_D over time for Target 2 (-25 dB)

Fig. 4 Probability of detection over pulses for Target 1 and 2 and with $\rho = 0.2$ and $K = 48$

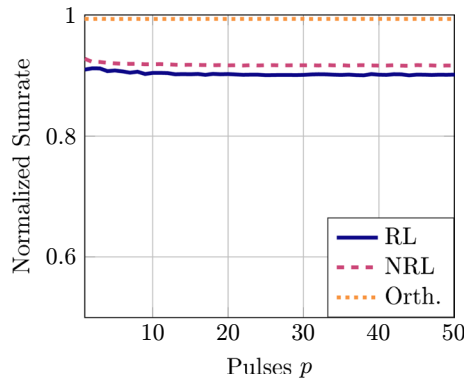


Fig. 5 Sum rate over time with $\rho = 0.2$, $K = 48$ and transmit SNR of 12 dB

they exhibit the lowest SNR values and thus represent the most challenging detection cases within the environment. As observed, the proposed RL-based method consistently outperforms both benchmark schemes. Specifically, neither the orthogonal beam pattern nor the NRL approach is able to detect Target 1. While the NRL method succeeds in detecting Target 2, it does this slower compared to the RL framework. The orthogonal beam pattern approach attains a maximum detection probability of only ~ 0.3 and lacks adaptability, as it does not incorporate any adaptive mechanism.

Figure 5 illustrates the normalized sum rate with respect to the zero-MUI benchmark over the time. As anticipated, the orthogonal beamforming configuration achieves the highest sum rate, owing to its absence of inter-beam interference. In contrast, both the RL and NRL schemes generate beams directed toward distinct spatial regions, thereby increasing MUI and reducing the overall throughput. It is further observed that the sum rate remains relatively stable over time across all approaches. However, in the proposed RL-based method, a transient higher sum rate is observed during the initial pulses, corresponding to the learning phase in which the agent explores and adapts to the target locations.

3.3 Impact of sensing–communication trade-offs on system performance

In the following analysis, the performance of the cognitive ISAC framework is evaluated under various sensing–communication trade-offs. The targets' angular position, SNRs, and the number of communication users remain identical to those defined in Sect. 3.2. In the following, we additionally investigate the sum rate performance for different communication SNR levels.

Figure 6a depicts the temporal evolution of the detection probability for different trade-off factors ρ . Only Target 1 is considered, as it exhibits the lowest SNR and thus represents the most challenging detection scenario. As expected, smaller values of ρ result in improved detection performance, as they assign greater emphasis to sensing optimization. Notably, for $\rho = 0.2$ and $\rho = 0.4$, the detection probabilities are nearly identical, with $\rho = 0.4$ exhibiting only a marginally lower performance. Figure 6b presents the average per-user sum rate across varying communication SNR levels. For low SNRs, the sum rate remains comparable across all trade-off factors. However, beyond 6 dB, the configuration with $\rho = 0.2$ begins to lag behind the others. The remaining trade-off settings yield similar performance up to approximately 12 dB, where a slight degradation is observed for $\rho = 0.4$. Overall, these results indicate that a trade-off factor of $\rho = 0.4$ provides a balanced compromise, offering strong sensing performance while maintaining a competitive communication sum rate.

3.4 Effect of the number of communication users on detection and sum rate

Up to this point, the analysis has been conducted assuming $K = 48$ communication users. In the following, we examine how system performance changes when the number of users is reduced. Specifically, we compare the detection probability and the sum rate for both high and low user densities. The targets' angular position and SNR values are again identical to those listed in Table 2. For the subsequent results, we set the trade-off factor to $\rho = 0.8$, thereby prioritizing communication performance during waveform optimization. As shown in Fig. 7a, a lower number of communication users yields improved detection probability compared to the high-user case. This outcome is expected, as mitigating MUI becomes increasingly challenging when more users are

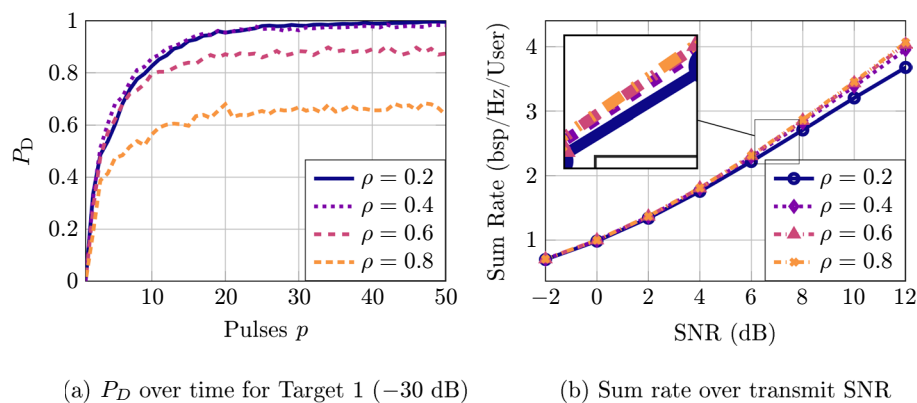


Fig. 6 Probability of detection over pulses and sum rate over varying transmit SNRs for different trade-off ρ

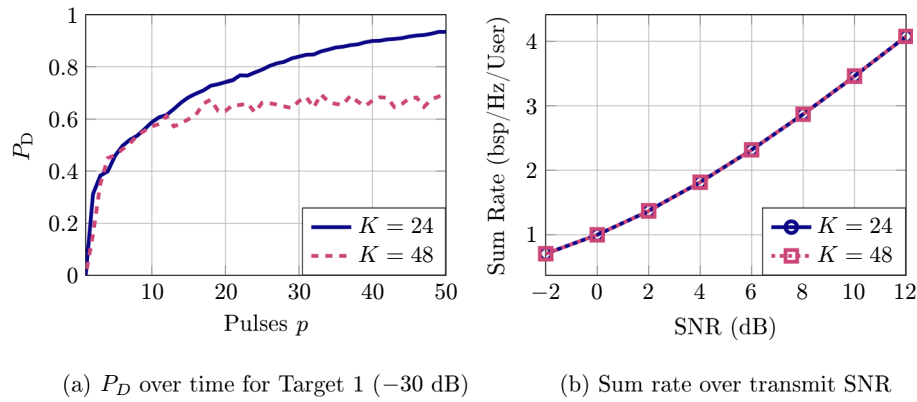


Fig. 7 Probability of detection over pulses and sum rate over varying transmit SNRs for different number of user K

Table 3 Dynamic Scenario

Target	Time interval					
	[1, 50]	[51, 100]	[101, 110]	[111, 120]	[121, 130]	[131, 140]
1	$(-0.4, -0.4)^1$ -30 dB ²	$(-0.4, -0.4)$ -30 dB	$(-0.4, -0.4)$ -31 dB	$(-0.4, -0.4)$ -32 dB	$(-0.4, -0.4)$ -33 dB	$(-0.4, -0.4)$ -34 dB
2	(0, 0) -25 dB	Absent -	Absent -	Absent -	Absent -	Absent -
3	Absent -	(0.3, 0.1) -30 dB	(0.3, 0.1) -30 dB	(0.3, 0.1) -30 dB	(0.3, 0.1) -30 dB	(0.3, 0.1) -30 dB

¹ Target's position

² Target's SNR

present. The average per-user sum rate, illustrated in Fig. 7b, remains approximately constant across different user counts. This observation indicates that the optimization framework effectively maintains a consistent communication rate per user, regardless of system load.

3.5 Performance evaluation in a dynamic sensing environment

The analyses presented in the previous sections were confined to a stationary sensing environment. We now extend the evaluation to a dynamic scenario to investigate system behavior under time-varying conditions. In this case, three targets are considered throughout the simulation, as summarized in Table 3. Initially, only Target 1 is present.

After 50 time steps, Target 1 disappears and Target 2 emerges with an SNR of -30 dB. Subsequently, beginning at time step 100, the SNR of Target 2 gradually decreases. The number of communication users is maintained at $K = 48$, and the following results are analyzed for multiple sensing–communication trade-off factors.

The evolution of the detection probability over time for all three targets is depicted in Fig. 8. The vertical black dashed lines indicate the time instances at which the environment undergoes changes. It can be observed that, for Target 1, the detection probability

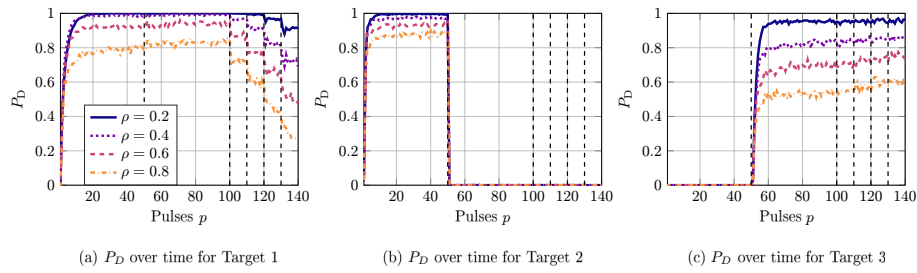


Fig. 8 Probability of detection over pulses for Target 1,2 and 3 with $K = 48$ and varying ρ

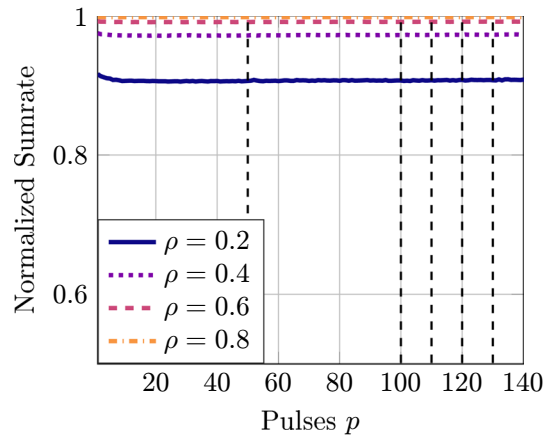


Fig. 9 Sum rate over time with $K = 48$ and transmit SNR of 12 dB and varying ρ

progressively decreases as ρ increases and the corresponding SNR diminishes. Moreover, the RL-based approach exhibits strong performance in detecting targets that appear at fixed spatial locations.

The corresponding average sum rate over time is presented in Fig. 9. The results show that the sum rate remains largely stable, except for an initial transient phase during which the RL agent is still adapting to the target positions, resulting in a slightly elevated sum rate.

3.6 Impact of sequential target appearance on communication performance

The absence of variation in the sum rate observed in the previous subsection, despite the dynamic environment, can be attributed to the constant number of detected targets in that scenario. To further investigate the system, we now consider a dynamic setting in which new targets are sequentially appearing over time. Specifically, a new target appears every 20 time steps, beginning with a single target and reaching a total of seven targets after 120 time steps. As before, the performance is compared across different trade-off factors ρ , with the number of communication users fixed at $K = 48$.

For a trade-off factor of $\rho = 0.2$, Fig. 10 clearly shows that the sum rate decreases each time a new target is introduced. This trend is intuitive, as the addition of more

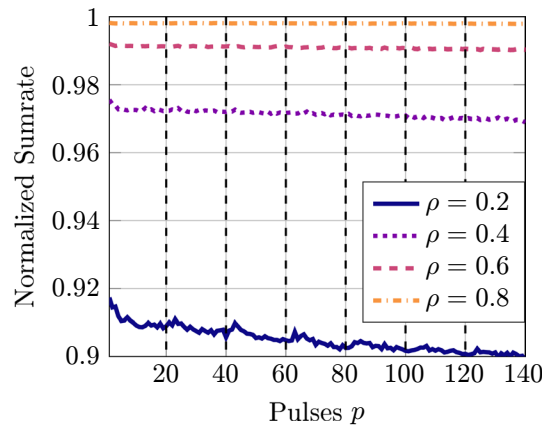


Fig. 10 Sum Rate over pulses for $K = 48$, 12 dB transmit SNR, and varying ρ with targets appearing every 20 pulses

targets increases the number of beams required for sensing, thereby reducing the optimization's ability to effectively suppress MUI when sensing performance is prioritized. In addition, the sum rate exhibits noticeable temporal instability. A similar decline in the sum rate is observed for $\rho = 0.4$. In contrast, for more communication-oriented configurations with $\rho = 0.6$ and $\rho = 0.8$, this effect is largely negligible.

4 Conclusions

This paper presented an RL-aided cognitive ISAC framework for massive MIMO systems employing a UPA configuration. The proposed approach integrates reinforcement learning into the cognitive radar perception–action cycle to enable adaptive waveform optimization under uncertain and dynamic environments. A Wald-type detector was utilized for robust target detection in non-Gaussian clutter, while a SARSA-based RL agent continuously refined its decision policy to maximize detection performance without prior environmental knowledge.

The proposed design jointly optimized radar sensing and downlink communication by formulating a unified waveform optimization problem that balances the conflicting objectives of sensing accuracy and communication throughput. Through an analytically derived closed-form solution, the framework provided a flexible and computationally efficient method to adjust this trade-off via a single weighting factor. As a result, the system achieved an adaptive equilibrium between multi-user interference suppression and beam pattern fidelity.

Comprehensive simulations confirmed that the RL-driven ISAC scheme consistently outperforms both orthogonal and non-learning adaptive baselines, particularly in low-SNR and dynamically varying environments. The results demonstrated that the cognitive learning mechanism enhances detection probability while maintaining stable and competitive communication rates. Furthermore, the study revealed that appropriate trade-off tuning enables the system to maintain balanced performance across varying user densities and channel conditions.

Overall, the key contribution of this work is a closed-loop cognitive coupling of robust detection, RL-driven 2D target-support learning, and analytical ISAC

waveform optimization, together with a quantitative evaluation of the resulting detection–sum-rate trade-offs. The presented framework establishes a robust and adaptive foundation for intelligent ISAC operation in next-generation wireless networks. It demonstrates the feasibility of radar–communication coexistence without explicit environmental modeling.

Robust disturbance covariance estimation

In general, the disturbance covariance matrix Γ is unknown. In the context of massive MIMO radar [22], however, it can be estimated directly from the *single* snapshot under test by means of robust techniques for dependent data. The key assumption required for consistent estimation is stated below.

Assumption 1 Let c denote a discrete-time, complex circular, possibly non-Gaussian random process $\{c_n, \forall n\}$ whose autocorrelation function decays polynomially. Formally,

$$r_c[m] \triangleq \mathbb{E}\{c_n c_{n-m}^*\} = \mathcal{O}(|m|^{-\gamma}), \quad m \in \mathbb{Z}, \quad \gamma > \frac{\epsilon}{\epsilon - 1}, \quad \epsilon > 1. \quad (\text{A1})$$

Assumption 1 is general enough to capture a wide range of disturbance models, including autoregressive–moving-average (ARMA) processes (possibly non-Gaussian) and compound Gaussian models. Moreover, it extends to higher-dimensional cases (e.g., 2D AR fields), provided that stationarity and polynomial correlation decay hold and the process is suitably vectorized.

Sample covariance estimation

Under Assumption 1, the disturbance covariance matrix can be consistently estimated by truncating the empirical covariance at a finite lag l . The entrywise estimator [22, Remark 1] is defined as

$$[\hat{\Gamma}_l]_{ij} = \begin{cases} \hat{c}_i \hat{c}_j^*, & j - i \leq l, \\ \hat{c}_i^* \hat{c}_j, & i - j \leq l, \\ 0, & |i - j| > l, \end{cases} \quad (\text{A2})$$

where $\hat{c}_n = y_n - \hat{\alpha} h_n$ denotes the residual disturbance after signal subtraction. The truncation parameter l must grow more slowly than $N^{1/3}$ to guarantee consistency [22, 37], where $N = N_t N_r$ is the number of *virtual* spatial antenna channels.

Asymptotic distribution of the test statistic

If Assumption 1 holds, Theorem 3 in [22] establishes the following asymptotic results for the Wald-type test statistic $\Lambda_{p,m}$:

- (i) Under the null hypothesis $\mathcal{H}_0$,

$$\Lambda_{p,m}(\mathbf{y}_{p,m} \mid \mathcal{H}_0) \underset{N \rightarrow \infty}{\overset{d}{\rightsquigarrow}} \chi_2^2(0). \quad (\text{A3})$$

Thus, the false-alarm rate is constant false alarm rate (CFAR) in the large-system regime, even without secondary data.

(ii) Under the alternative hypothesis $\mathcal{H}_1$,

$$\Lambda_{p,m}(\mathbf{y}_{p,m} | \mathcal{H}_1) \underset{N \rightarrow \infty}{\overset{d}{\sim}} \chi_2^2(\zeta), \quad (\text{A4})$$

with non-centrality parameter

$$\zeta = 2 |\alpha|^2 \frac{\|\mathbf{h}\|^4}{\mathbf{h}^H \Gamma \mathbf{h}}. \quad (\text{A5})$$

Detection probability in the large-system regime

In the asymptotic regime ($N \rightarrow \infty$), the probability of detection at threshold η converges to

$$P_D(\eta) \xrightarrow{N \rightarrow \infty} Q_1(\sqrt{\zeta}, \sqrt{\eta}), \quad (\text{A6})$$

where $Q_1(\cdot, \cdot)$ denotes the first-order Marcum Q -function [34]. Once the nominal false-alarm probability P_{FA} is fixed, the threshold η follows accordingly, and P_D becomes a closed-form function of ζ . Importantly, ζ depends on the disturbance only through the ratio $\|\mathbf{h}\|^4 / (\mathbf{h}^H \Gamma \mathbf{h})$. Hence, it is sufficient that $\hat{\Gamma}_l$ consistently estimates Γ in this ratio, even if the exact distribution of the disturbance remains unknown.

Discussion

By leveraging robust estimation theory for dependent observations [38] and the large antenna dimension, the Wald-type detector achieves invariance to the disturbance distribution. Consequently, it enjoys a strong CFAR property without requiring secondary (training) snapshots from homogeneous ranges.

Solution for the trade-off design optimization problem

To solve $\mathcal{P}_{0,1}$ a low-complexity algorithm is derived in [17] that achieves the global optimum in the following. Let us further expand the objective function of in $\mathcal{P}_{0,1}$ as

$$\begin{aligned} \|\mathbf{A}\mathbf{X} - \mathbf{B}\|_F^2 &= \text{tr}((\mathbf{A}\mathbf{X} - \mathbf{B})^H(\mathbf{A}\mathbf{X} - \mathbf{B})) \\ &= \text{tr}(\mathbf{X}^H \mathbf{A}^H \mathbf{A} \mathbf{X}) - \text{tr}(\mathbf{X}^H \mathbf{A}^H \mathbf{B}) - \text{tr}(\mathbf{B}^H \mathbf{A} \mathbf{X}) + \text{tr}(\mathbf{B}^H \mathbf{B}). \end{aligned} \quad (\text{B7})$$

Defining $\mathbf{Q} = \mathbf{A}^H \mathbf{A}$ and $\mathbf{G} = \mathbf{A}^H \mathbf{B}$, the problem can be rewritten as

$$\min_{\mathbf{X}} \text{tr}(\mathbf{X}^H \mathbf{Q} \mathbf{X}) - 2 \text{Re}(\text{tr}(\mathbf{X}^H \mathbf{G})) \quad (\text{B8})$$

$$\text{s.t. } \|\mathbf{X}\|_F^2 = LP_T. \quad (\text{B9})$$

Since $\mathbf{Q}$ is Hermitian, (B8) is the matrix version of the trust-region subproblem (TRS), for which strong duality holds. The Lagrangian is

$$\mathcal{L}(\mathbf{X}, \lambda) = \text{tr}(\mathbf{X}^H \mathbf{Q} \mathbf{X}) - 2 \text{Re}(\text{tr}(\mathbf{X}^H \mathbf{G})) + \lambda (\|\mathbf{X}\|_F^2 - LP_T), \quad (\text{B10})$$

where λ is the dual variable. Let $\mathbf{X}_{\text{opt}}$ and λ_{opt} be primal/dual optima. The optimality conditions are

$$\nabla_{\mathbf{X}} \mathcal{L}(\mathbf{X}_{\text{opt}}, \lambda_{\text{opt}}) = 2(\mathbf{Q} + \lambda_{\text{opt}} \mathbf{I}_N) \mathbf{X}_{\text{opt}} - 2\mathbf{G} = \mathbf{0}, \quad (\text{B11a})$$

$$\|\mathbf{X}_{\text{opt}}\|_F^2 = LP_T, \quad (\text{B11b})$$

$$\mathbf{Q} + \lambda_{\text{opt}} \mathbf{I}_N \succeq \mathbf{0}, \quad (\text{B11c})$$

which yield

$$\mathbf{X}_{\text{opt}} = (\mathbf{Q} + \lambda_{\text{opt}} \mathbf{I}_N)^{\dagger} \mathbf{G}. \quad (\text{B12})$$

Using (B11b) and (B11c) we have

$$\left\| (\mathbf{Q} + \lambda_{\text{opt}} \mathbf{I}_N)^{\dagger} \mathbf{G} \right\|_F^2 = \left\| \mathbf{V}(\mathbf{\Lambda} + \lambda_{\text{opt}} \mathbf{I}_N)^{-1} \mathbf{V}^H \mathbf{G} \right\|_F^2 = LP_T, \quad \lambda_{\text{opt}} \geq -\lambda_{\min}, \quad (\text{B13})$$

where $\mathbf{Q} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H$ is the eigen-decomposition with $\mathbf{V}$ unitary and diagonal $\mathbf{\Lambda}$, and $\lambda_{\min}$ is the minimum eigenvalue of $\mathbf{Q}$. Define

$$P(\lambda) = \left\| \mathbf{V}(\mathbf{\Lambda} + \lambda \mathbf{I}_N)^{-1} \mathbf{V}^H \mathbf{G} \right\|_F^2 = \sum_{i=1}^N \sum_{j=1}^L \frac{(|\mathbf{V}^H \mathbf{G}|_{ij})^2}{(\lambda + \lambda_i)^2}, \quad (\text{B14})$$

which is strictly decreasing and convex for $\lambda \geq -\lambda_{\min}$. Hence, λ_{opt} can be found efficiently by a 1-D line search (e.g., Golden-section), after which $\mathbf{X}_{\text{opt}}$ follows from (B12). For clarity, we summarize the above approach in Algorithm 2.

Algorithm 2 Low-complexity Algorithm

Input: $\mathbf{H}, \mathbf{S}, \mathbf{X}_0$, weight $0 \leq \rho \leq 1$, P_T .

Output: Global minimizer $\mathbf{X}_{\text{opt}}$.

1. Form $\mathbf{A}, \mathbf{B}$ as in (18) and compute $\mathbf{Q} = \mathbf{A}^H \mathbf{A}$, $\mathbf{G} = \mathbf{A}^H \mathbf{B}$.
 2. Eigendecompose $\mathbf{Q} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H$ and set the search interval $[-\lambda_{\min}, b]$ with $b \geq 0$.
 3. Find λ_{opt} via line search (e.g., Golden-section).
 4. Set $\mathbf{X}_{\text{opt}} = (\mathbf{Q} + \lambda_{\text{opt}} \mathbf{I}_N)^{\dagger} \mathbf{G}$.
-

Solution for the radar waveform optimization problem

Given a target covariance matrix $\mathbf{R}_d$ corresponding to a well-designed MIMO radar beam pattern, the MUI minimization problem can be formulated as

$$\begin{aligned} \min_{\mathbf{X}_0} \quad & \|\mathbf{H} \mathbf{X}_0 - \mathbf{S}\|_F^2 \\ \text{s.t.} \quad & \frac{1}{L} \mathbf{X}_0 \mathbf{X}_0^H = \mathbf{R}_d. \end{aligned} \quad (\text{C15})$$

The matrix $\mathbf{R}_d$ is Hermitian and positive semidefinite.

We consider the Cholesky factorization

$$\mathbf{R}_d = \mathbf{F}\mathbf{F}^H, \quad (\text{C16})$$

where $\mathbf{F} \in \mathbb{C}^{N \times N}$ is lower triangular. Without loss of generality, we assume $\mathbf{R}_d$ is positive definite, which guarantees that $\mathbf{F}$ is invertible. Substituting this decomposition into the constraint yields

$$\frac{1}{L} \mathbf{F}^{-1} \mathbf{X}_0 \mathbf{X}_0^H \mathbf{F}^{-H} = \mathbf{I}_N. \quad (\text{C17})$$

Defining the normalized variable

$$\tilde{\mathbf{X}}_0 \triangleq \sqrt{\frac{1}{L}} \mathbf{F}^{-1} \mathbf{X}_0, \quad (\text{C18})$$

the problem reduces to

$$\begin{aligned} \min_{\tilde{\mathbf{X}}_0} \quad & \left\| \sqrt{L} \mathbf{H} \mathbf{F} \tilde{\mathbf{X}}_0 - \mathbf{S} \right\|_F^2 \\ \text{s.t.} \quad & \tilde{\mathbf{X}}_0 \tilde{\mathbf{X}}_0^H = \mathbf{I}_N. \end{aligned} \quad (\text{C19})$$

This is recognized as an orthogonal Procrustes problem (OPP). Its globally optimal solution is

$$\tilde{\mathbf{X}}_0 = \tilde{\mathbf{U}} \mathbf{I}_{N \times L} \tilde{\mathbf{V}}^H, \quad (\text{C20})$$

where $\tilde{\mathbf{U}} \tilde{\mathbf{\Sigma}} \tilde{\mathbf{V}}^H = \mathbf{F}^H \mathbf{H}^H \mathbf{S}$ is the singular value decomposition (SVD) of $\mathbf{F}^H \mathbf{H}^H \mathbf{S}$.

Finally, substituting back, the optimal transmit waveform matrix is obtained as

$$\mathbf{X}_0 = \sqrt{L} \mathbf{F} \tilde{\mathbf{U}} \mathbf{I}_{N \times L} \tilde{\mathbf{V}}^H. \quad (\text{C21})$$

Abbreviations

2D	Two-dimensional
6G	Sixth-generation
AR	Autoregressive
ARMA	Autoregressive–moving-average
AWGN	Additive white Gaussian noise
BS	Base station
CFAR	Constant false alarm rate
CR	Cognitive Radar
GLRT	Generalized likelihood ratio test
ISAC	Integrated sensing and communication
MIMO	Multiple-input and multiple-output
MUI	Multi-user interference
NRL	Non reinforcement learning
PSD	Power spectral density
QCQP	Quadratically constrained quadratic program
RL	Reinforcement learning
SARSA	State–action–reward–state–action
SDP	Semidefinite Program
SDR	Semidefinite Relaxation
SER	Symbol error rate
SINR	Signal-to-interference-plus-noise ratio
SNR	Signal-to-noise ratio
STAP	Space–time adaptive processing
SVD	Singular value decomposition
TRS	Trust-region subproblem

ULA	Uniform linear array
UPA	Uniform planar array
QPSK	Quadrature phase-shift keying

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Author Contributions

AU conceived the study, implemented it, conducted the analyses, and wrote the manuscript. All authors were involved in idea development and manuscript review, and all approved the final manuscript.

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Data Availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest

The authors declare that they have no conflict of interest.

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